

CBCS SCHEME

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BCS402

Fourth Semester B.E./B.Tech. Degree Examination, Dec.2025/Jan.2026 Microcontrollers

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Mention the difference between : i) Microprocessor and Microcontroller ii) CISC and RISC architectures	8	L2	CO1
	b.	With a neat diagram, explain ARM core dataflow model.	6	L2	CO1
	c.	With a neat diagram, explain embedded system hardware.	6	L2	CO1
OR					
Q.2	a.	Explain the various modes of operation of ARM processor and banked registers.	8	L2	CO1
	b.	What is a pipeline? With neat diagram explain the various blocks in a 3-stage pipeline of ARM processor organization.	6	L2	CO1
	c.	Discuss the various fields in CPSR with neat sketch.	6	L2	CO1
Module – 2					
Q.3	a.	With example illustrate how following instructions work. i) MLA ii) MUL iii) SMLAL iv) UMULL.	8	L3	CO2
	b.	Explain Single register load store addressing mode syntax, table index mode with an example.	8	L2	CO2
	c.	Explain Barrel shifter operation in ARM processor with neat diagram. If $r_5 = 5$, $r_7 = 8$ using the following instructions, write values of r_5 , r_7 after execution of MOV r_7, r_5 , LSL # 2.	4	L2	CO2
OR					
Q.4	a.	Along with suitable examples describe various logical and comparison instructions.	8	L2	CO2
	b.	Discuss the Branch instructions and SWAP instructions with example.	8	L2	CO2
	c.	Explain briefly co-processor instructions of ARM processor.	4	L2	CO2
Module – 3					
Q.5	a.	Explain basic C-Data types with example codes.	10	L2	CO3
	b.	Discuss how Registers are allocated to optimize the program.	10	L2	CO3

OR					
Q.6	a.	Explain C looping structures with example codes.	10	L2	CO3
	b.	Explain function calls and pointer Aliasing with example codes.	10	L2	CO3
Module – 4					
Q.7	a.	With a neat sketch, explain exceptions and associated modes.	10	L2	CO4
	b.	With the help of vector table, explain processor modes. Also mention the exception priority levels.	10	L2	CO4
OR					
Q.8	a.	Explain interrupt latency, IRQ and FIQ exceptions in detail with neat sketches.	10	L2	CO4
	b.	Explain Firmware and bootloader, with sand stone example, explain detail directory layout.	10	L2	CO4
Module – 5					
Q.9	a.	With neat diagram, explain the relationship of cache between processor core and main memory.	8	L2	CO5
	b.	With neat sketch, explain set associatively.	8	L2	CO5
	c.	Explain logical and physical cache with neat diagram.	4	L2	CO5
OR					
Q.10	a.	Briefly, explain cache policies.	8	L2	CO5
	b.	Explain direct mapped cache and thrashing.	8	L2	CO5
	c.	Write short note on co-processor 15.	4	L2	CO5
